Generalized Gamma Convolution Laws from Continuous Branching Processes

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Continuous-State Branching Processes on $[0,\infty)$

- $(X_t)_{t\geq 0}$ = CSBP if $\mathbb{P}_{x+y}(X_t\in\cdot)=\mathbb{P}_x(X_t\in\cdot)*\mathbb{P}_y(X_t\in\cdot),\ \ x,y,t\geq 0.$

- Chapman–Kolmogorov:
$$\mathbb{E}_x[e^{-\lambda X_t}]=e^{-x\,u_t(\lambda)},\ \ \lambda\geq 0\Longrightarrow u_t\big(u_s(\lambda)\big)=u_{t+s}(\lambda).$$

- Regularity condition and branching mechanism (Biggins, Bingham, Grey, Silverstein, 70–80's):
$$\lim_{t\to 0}\frac{u_t(\lambda)-\lambda}{t}=c(\lambda)\Longrightarrow u_0(\lambda)=\lambda,\ \ \frac{\partial}{\partial t}u_t(\lambda)=c\big(u_t(\lambda)\big)$$
$$c(\lambda)=\alpha+\beta\lambda-\gamma\lambda^2+\int_{(0,\infty)}\big(1-e^{-\lambda x}-\lambda x\,e^{-x}\big)\,Q(dx).$$
$\alpha,\gamma\geq 0,\ \beta\in\mathbb{R},\ Q=$ Lévy measure.

Motivation: Branching Equation

$$\boxed{\text{Supercritical case: } \gamma = 0 \text{ and } c'(0+) \in (0, \infty).}$$

$$c(\lambda) = k + \delta\lambda + \int_{(0,\infty)} (1 - e^{-\lambda x}) Q(x)\, dx \in \mathcal{BF}.$$

- The Lamperti transform of $(X_t)_{t \geq 0}$ involves

$$(Y_t)_{t\geq 0} = \text{subordinator with Bernstein function } c$$

- (Biggins, Bingham, Grey 70–80's): $\exists$ a normalization $\eta(t)$ s.t. $e^{-\eta(t)\, X_t}$ = converging positive martingale, hence

$$W_t = \eta(t)\, X_t \xrightarrow{a.s.} W_\infty \sim \mathrm{ID}(\mathbb{R}_+).$$

- The branching equation: the Bernstein function w of W_∞ satisfies

$$\boxed{w(0) = 0, \quad \lambda\, w'(\lambda) = c(w(\lambda)), \;\; \lambda \geq 0 \quad \textbf{(BE)}.}$$

Pakes' Conjecture (2018)

$\boxed{\text{Pakes' conjecture: } W_\infty \sim \mathrm{BO} \text{ or } \mathrm{GGC} \Longleftrightarrow w \in \mathcal{CBF} \text{ or } \mathcal{TBF}.}$

The central equation is

$$\boxed{\lambda\, w'(\lambda) = c(w(\lambda)), \quad \lambda \geq 0 \quad \textbf{(BE)}.}$$

Key quantity: $$\tau(\lambda) = \int_0^\lambda \left(\frac{1}{c(x)} - \frac{1}{x} \right) dx, \quad c \in \mathcal{BF}.$$

Pakes (2018): under a log-moment condition for Q,

- $c \in \mathcal{SB} \Longleftrightarrow \tau(\lambda),\ \lambda\tau'(\lambda) \in \mathcal{BF} \Longrightarrow w \in \mathcal{SB}$.
- $c \in \mathcal{CBF} \Longleftrightarrow \tau \in \mathcal{TBF}$.

Motivation by Examples

i	$c_i(\lambda)$	$\tau_i(\lambda)$	$w_i(\lambda)$	$w_i \in \mathcal{TBF}$?
1	$1 - e^{-\lambda}$	$\log\left(\frac{e^\lambda - 1}{\lambda}\right)$	$\log(1+\lambda)$	$\checkmark$
2	$\alpha\lambda$	$\nexists$	λ^α	$\checkmark$ for $\alpha \in (0,1]$
3	$\delta(\lambda + 1 - (1+\lambda)^{1-\frac{1}{\delta}})$	$\log\left(\frac{(1+\lambda)^{1/\delta} - 1}{\lambda}\right)$	$(1+\lambda)^\delta - 1$	$\checkmark$ for $\delta \in (0,1]$
4	$\frac{\lambda}{1+\lambda}$	λ	Lambert func.	$\checkmark$

Note that

- Concavity is necessary for Bernstein structure for w. Solutions may exist outside $\mathcal{BF}$:

$$c_2(\lambda) = \alpha\lambda \Longrightarrow w_2 \text{ is convex if } \alpha > 1.$$

- The Lambert function w_4 has no closed form, it satisfies

$$w_4(\lambda) = \lambda\, e^{-w_4(\lambda)} \in \mathcal{TBF} \quad \text{(Pakes 2011)}.$$

Intuition and approach to solve Pakes' conjecture

- Our analysis is performed WITHOUT relying on branching process arguments. It illustrate that the *generalized Lambert functions* solutions of

$$\boxed{w(\lambda)e^{\tau(w(\lambda))} = \lambda}$$

are often Thorin Bernstein functions.

- From last tabular: the condition $c'(0) \leq 1$ appears essential !!
- $c \in \mathcal{CBF} \Longrightarrow$ there is no solution w of **(BE)** in $\mathcal{CBF} \setminus \mathcal{TBF}$.
- Another Key quantity is the inverse function $\Psi = w^{-1}$:

$$\textbf{(BE)} \iff \Psi(\lambda) = \Psi'(\lambda)\, c(\lambda).$$

When is Ψ a "nice" Lévy-Laplace exponent of the Spectrally Negative type ?

- Important fact:

$$\boxed{\theta > 1 \Longrightarrow c_\theta(x) := \theta\, x^{1-\frac{1}{\theta}}\, c(x^{\frac{1}{\theta}}) \in \mathcal{BF}\ (\in \mathcal{CBF} \text{ if } \theta \geq 2).}$$

Solutions for the (BE)

$$\boxed{\lambda\, w'(\lambda) = c\big(w(\lambda)\big), \qquad c(\lambda) = k + \delta\lambda + \int_{(0,\infty)} (1 - e^{-\lambda x}) Q(x)\, dx \quad \textbf{(BE)}.}$$

$$c'(0+) = 1 \quad \text{and} \quad \int_1^\infty x\ \log x\ Q(dx) < \infty \ \ \textbf{(LMC)}.$$

Theorem 1 (existence and unicity result):

1) $c(0) = 0 \iff \exists\ w$ solution of **(BE)** unique up to scaling: $w(\rho\lambda),\quad \rho > 0$.

2) $\mathbf{c'(0+) \le 1} = \delta = 1 \iff c(\lambda) = \lambda$, and $w(\lambda) = \rho\,\lambda,\ \lambda \ge 0$, for some $\rho > 0$.

3) $\mathbf{c'(0+) \le 1},\ \delta < 1 \iff w$ concave $\iff w(\lambda) = \int_0^\infty (1 - e^{-\lambda x}) \frac{k(x)}{x}\, dx,\ \ k \searrow$.

4) $w'(0+) = \begin{cases} \infty, & \text{if} \quad \mathbf{c'(0+) < 1}; \\ 1, & \text{if} \ \ \textbf{(LMC)} \ \ \text{holds.} \end{cases}$

5) **(LMC)** $\iff \tau(\lambda) = \int_0^\lambda \left(\frac{1}{c(x)} - \frac{1}{x} \right) dx \in [0, \infty),\ \forall \lambda \ge 0 \Longrightarrow w \in \mathcal{SB}$ and

$$w_*(\lambda) := \frac{\lambda}{w(\lambda)} = e^{\tau(w(\lambda))} \quad \text{(generalized Lambert functions).}$$

Reminder on ID

- $f \in \mathcal{CM} \iff (-1)^k f^{(k)}(\lambda) \geq 0 \overset{\text{Bernstein}}{\iff} f(\lambda) = \int_{[0,\infty)} e^{-\lambda x}\mu(dx),\ \lambda > 0.$
- $\phi \in \mathcal{BF} \iff \phi \geq 0,\ \phi' \in \mathcal{CM}$, i.e.
$$\phi(\lambda) = a + b\lambda + \int_{(0,\infty)} (1 - e^{-\lambda x})\,\pi(dx),\ \ \lambda \geq 0,\ \ (\mathrm{a},\ \mathrm{b} \geq 0).$$
- Lévy exponent
$$\Psi(u) = \mathrm{i}ud - \frac{c^2u^2}{2} + \int_{\mathbb{R}\setminus\{0\}} \left(e^{\mathrm{i}ux} - 1 - \mathrm{i}uh(x)\right)\pi(dx),\ \ u \in \mathbb{R}.$$
- Lévy Processes: $t > s \geq 0 \Longrightarrow Z_t - Z_s \perp\!\!\!\perp \mathcal{F}_s$ and $Z_t - Z_s \overset{\mathrm{d}}{=} Z_{t-s}$.
- Subordinator: η_t = nondecreasing Lévy process.
- Lévy–Khintchine Formulae: $\mathbb{E}[e^{\mathrm{i}\,u\,Z_t}] = e^{t\Psi(u)}, \quad \mathbb{E}[e^{-\lambda\,\eta_t}] = e^{-t\phi(\lambda)}.$
- Infinite divisibility: $X \sim \mathrm{ID}(\mathbb{R})$ (resp $\mathrm{ID}(\mathbb{R}_+)$) $\iff X \overset{\mathrm{d}}{=} Z_1$ (resp $X \overset{\mathrm{d}}{=} \eta_1$).

Important Subclasses of $\mathcal{BF}$

- Jurek Bernstein functions ($\mathcal{JBF}$): $\frac{\pi(dx)}{dx} \searrow$.
- Special Bernstein functions ($\mathcal{SB}$): ϕ and $\phi_*(\lambda) = \frac{\lambda}{\phi(\lambda)} \in \mathcal{BF}$.
- Complete Bernstein functions of order $a > -1$ ($\mathcal{CBF}_a$):

$$\phi \in \mathcal{BF} \text{ and } \frac{\pi(dx)}{dx} = x^{a-1} \int_{(0,\infty)} e^{-x\,u} \Delta(du).$$

Equivalently, $\phi \geq 0$ and $\phi' \in \mathcal{S}_a = \left\{ b + \int_{(0,\infty)} \frac{1}{(\lambda+u)^a} \, \Delta(du) \right\}$.

Examples: $\alpha \in (0,1) \Longrightarrow \lambda^\alpha \in \mathcal{CBF}_{-\alpha}$, $\lambda^\alpha/(1+\lambda^\alpha) \in \mathcal{CBF}_\alpha$.

- Complete Bernstein functions ($\mathcal{CBF} \subset \mathcal{SB}$):

$$\mathcal{CBF} = \mathcal{CBF}_1 \longleftrightarrow \mathrm{BO} = \overline{\bigcap \text{mixtures of exponentials.}}$$

- Thorin Bernstein functions ($\mathcal{TBF}$):

$$\mathcal{TBF} = \mathcal{CBF}_0 \longleftrightarrow \mathrm{GGC} = \overline{\bigcap \text{ mixture of gamma distributions.}}$$

Lévy–Laplace exponents

- Lévy–Laplace exponents ($\mathcal{LE}$) $\longleftrightarrow$ SPLN:

$$\psi(\lambda) = \mathrm{a} + \mathrm{b}\lambda + \mathrm{c}\,\lambda^2 + \int_{(0,\infty)} (e^{-\lambda x} - 1 + \lambda x)\pi(dx), \quad \lambda \geq 0,$$

Proposition 1 (Complement to Bertoin, Roynette and M. Yor 2004): Let $\Psi : [0,\infty) \to [0,\infty)$ continuous , differentiable on $(0,\infty)$ s.t. $\Psi(0) = 0$. Define

$$\phi := \Psi^{-1},\ \phi_*(\lambda) = \lambda/\phi(\lambda) \text{ and } \Psi^*(\lambda) := \lambda\phi_*(\lambda) = \lambda^2/\phi(\lambda).$$

1) The following assertions are equivalent
   - (i) $\Psi \in \mathcal{LE}$;
   - (ii) $\phi \in \mathcal{BF}$, and $\phi_* = g(\phi)$, for some $g \in \mathcal{JBF}$;
   - (iii) ϕ is differentiable on $(0,\infty)$ and $1/\phi' = h(\phi)$, for some $h \in \mathcal{BF}$.

2) Under any of the latter, the following holds:
   - (i) $\phi \in \mathcal{SB}$ and $1/(\phi')^2 \in \mathcal{BF}$;
   - (ii) $\Psi' \in \mathcal{SB} \Longrightarrow \phi^2 \in \mathcal{LE}$;
   - (iii) The potential density of ϕ is convex $\Longleftrightarrow \phi_* \in \mathcal{JBF} \Longleftrightarrow \Psi^* \in \mathcal{LE}$.

Important Subclasses of $\mathcal{LE}$

- Complete Lévy-Laplace exponents of order $a > -2$ ($\mathcal{CLE}_a$):

$$\psi \in \mathcal{LE} \text{ and } \frac{\pi(dx)}{dx} = x^{a-2} \int_{(0,\infty)} e^{-x\,u} \Delta(du).$$

Equivalently, $\psi \geq 0$ and $\psi' \in \mathcal{CBF}_a$.

Proposition 2: We assume the formalism of Proposition 1.

1) The following assertions are equivalent:
   (i) $\Psi \in \mathcal{CLE}_2$;
   (ii) $\phi \in \mathcal{CBF}$ and $\phi_* = g(\phi)$, with some $g \in \mathcal{CBF}$ (hence $\Psi^* \in \mathcal{CLE}_2$);
   (iii) ϕ solves the differential equation $1/\phi' = h(\phi)$, with some $h \in \mathcal{CBF}_2$.
2) We have $(i) \Longrightarrow (ii) \Longleftrightarrow (iii) \Longrightarrow (iv)$:
   (i) $\lambda \mapsto \Psi(\lambda)/\lambda \in \mathcal{TBF}$;
   (ii) $\Psi \in \mathcal{CLE}_1$;
   (iii) $\phi' = s(\phi)$, for some $s \in \mathcal{S}_1$;
   (iv) $\phi \in \mathcal{TBF}$, $\phi^2 \in \mathcal{CLE}_1$ and $\phi_*^2 \in \mathcal{CBF}$.
3) If $\Psi \in \mathcal{CLE}_0$ (respectively $\mathcal{CLE}_1$), then $1/(\phi')^2 \in \mathcal{TBF}$ (respectively $\mathcal{CBF}$).

Infinite Divisibility Consequences

Corollary 1: Assume $\mathbf{c}'(\mathbf{0}+) \leq \mathbf{1}$.

1) We have

$$\lambda \mapsto e^{-tw(\lambda)}, \;\; e^{-t\lambda w'(\lambda)}, \;\; \left(\frac{w(\lambda)}{\lambda}\right)^t, \;\; (w'(\lambda))^t \in \mathcal{CM}, \;\; t \geq 0, \quad \textbf{(IDW)}$$

and

$$\psi(\lambda) := \lambda\, w(\lambda) \in \mathcal{LE} \text{ and } \varphi := \psi^{-1} \in \mathcal{SB}$$

satisfies the conjugacy equations

$$\lambda = \varphi\, w(\varphi), \quad \text{and} \quad \frac{1}{(\varphi')^2} = (w + c \circ w)^2(\varphi) \in \mathcal{BF}.$$

Stochastic interpretations for **(IDW)** *are too long to be developed here!*.

2) If in addition **(LMC)** holds, then

$$\Psi = w^{-1} \in \mathcal{LE} \iff e^{\tau} \in \mathcal{JBF} \iff e^{\tau(\lambda)}(1 + \lambda\tau'(\lambda)) \in \mathcal{BF}.$$

Answers to Pakes' Conjecture: $W_\infty \sim \mathrm{GGC}$

$$\boxed{\theta > 1 \Longrightarrow c_\theta(x) := \theta\, x^{1-\frac{1}{\theta}}\; c(x^{\frac{1}{\theta}}) \in \mathcal{BF}\ (\in \mathcal{CBF} \text{ if } \theta \geq 2)}$$

$$c_\theta'(0+) = \theta\, c'(0+), \quad \text{and} \;\; \delta_\theta = \;\textit{drift term of}\; c_\theta = \theta\, \delta.$$

Corollary 2 (Composition with power functions) Assume $\mathbf{c}'(\mathbf{0}+) \leq \mathbf{1}$.

1) For all $\sigma > 0$ and $\theta > 1$, the function $w_\sigma(\lambda) := w(\lambda^{\sigma)}$ and $w^\theta(\lambda)$ are the solution of

$$\lambda\, w_\sigma'(\lambda) = \sigma\, c\big(w_\sigma(\lambda)\big) \quad \text{and} \quad \lambda\, (w^\theta)'(\lambda) = c_\theta\big(w^\theta(\lambda)\big), \;\; \lambda > 0.$$

2) If $\sigma,\ \theta \leq 1/c'(0+)$, and if c replaced by the functions $\sigma \times c$ and c_θ respectively, then w_σ and w^θ satisfy 1) of Corollary 1. Moreover, if

$$\sigma = \theta = \frac{1}{c'(0+)} \;\text{ and }\; \int_1^\infty x\, \log x\; Q(dx) < \infty,$$

they satisfy 2) of Corollary 1.

Answers to Pakes' Conjecture: $W_\infty \sim \mathrm{GGC}$

Theorem 1: Let $a = c'(0+)$. Assume $a < \frac{1}{2}$, or $a = \frac{1}{2}$ and **(LMC)**. Then

$$w \in \mathcal{CBF}_a \text{ (i.e. Lévy density} = x^{a-1}\, l(x),\ l \in \mathcal{CM}) \implies w \in \mathcal{TBF}$$

Theorem 2: Assume $c'(0+) \leq 1$ and recall $\Psi = w^{-1}$. Then

1) $c \in \mathcal{CBF} \iff \tau' \in \mathcal{S}_1$, and under (LMC), $c \in \mathcal{CBF} \iff \tau \in \mathcal{TBF}$.
2) If $c \in \mathcal{CBF}$ and, either $c'(0+) < 1$ or **(LMC)**. Then

$$\delta \in [1/2, 1) \iff \Psi \in \mathcal{CLE}_2 \text{ (i.e. } \Psi \in \mathcal{LE} \text{ and its Lévy density} \in \mathcal{CM}).$$

Under the latter $w \in \mathcal{TBF}$.

Theorem 3: Assume $c \in \mathcal{BF}$ and **one** of the following

$$c'(0+) < \min\left(\frac{1}{2}, 2\delta\right), \text{ and } \theta \in \left[\max\left(\frac{1}{2\delta}, 2\right), \frac{1}{c'(0+)}\right),$$

$$c'(0+) \leq \frac{1}{2}, \quad \int_1^\infty x \log x\, Q(dx) < \infty, \text{ and } \theta = \frac{1}{c'(0+)}.$$

Then $\Psi(\lambda^{1/\theta}) \in \mathcal{CLE}_2$, $w^\theta \in \mathcal{TBF}$ and $w^\varkappa \in \mathcal{CBF}$ for all $0 < \varkappa \leq \theta \implies w \in \mathcal{TBF}$

Example: Perturbing the Lambert function

Solving **(BE)** with $c(\lambda) = \delta\lambda + \gamma\frac{\lambda}{1+\lambda}$, $\quad \delta,\ \gamma > 0$, we obtain

$$\Psi(\lambda) = w^{-1}(\lambda) = \lambda\varphi(\lambda), \quad \varphi\ (\lambda) = \lambda^{\frac{1}{\delta+\gamma}-1}\,(\delta\lambda+\delta+\gamma)^{\frac{\gamma}{\delta(\delta+\gamma)}}, \quad \lambda\ \geq 0.$$

We deduce that

$$\delta+\gamma = c'(0+) \leq 1 \ \text{ and } \ \frac{1}{2} \leq \delta < 1 \Longleftrightarrow g \in \mathcal{CBF} \Longleftrightarrow \Psi \in \mathcal{CLE}_2,$$

$$\textbf{(LMC)} \Longleftrightarrow \delta+\gamma = 1 \Longleftrightarrow \Psi(\lambda) = \lambda\,(\delta\lambda+1)^{\frac{1}{\delta}-1}.$$

With $\alpha = \frac{1}{\delta}-1 \in (0,1]$, $\ c_\alpha = \frac{\alpha}{(1+\alpha)^\alpha\ \Gamma(1-\alpha)}$,

$$\Psi(\lambda) = \lambda + \begin{cases} \dfrac{\lambda^2}{2}, & \text{if} \quad \delta = \frac{1}{2} \\ c_\alpha \displaystyle\int_0^\infty (e^{-\lambda x} - 1 + \lambda x)\, e^{-x}\, \frac{\alpha+1+x}{x^{\alpha+2}}\, dx, & \text{if} \quad \frac{1}{2} < \delta < 1. \end{cases}$$

- $\delta = 1/2 \Longrightarrow w(\lambda) = \Psi^{-1}(\lambda) = 2(\sqrt{(1+\lambda)} - 1) \in \mathcal{TBF}$.
- $1/2 < \delta < 1 \overset{\text{Theorem 3}}{\Longrightarrow} w = \Psi^{-1} \in \mathcal{TBF}$ but is not explicit!

Main Contributions and Extentions

1. Existence and uniqueness of Bernstein solutions.
2. Sharp criteria for complete Bernstein structure.
3. New Thorin Bernstein constructions.
4. Stability under powers and transformations.
5. New links between CBP equations and GGC laws: $W_\infty \sim \mathrm{GGC}$
6. Extend the study for $c \in \mathcal{LE}$.

Dziękuję!, Thank You!

Selected References

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